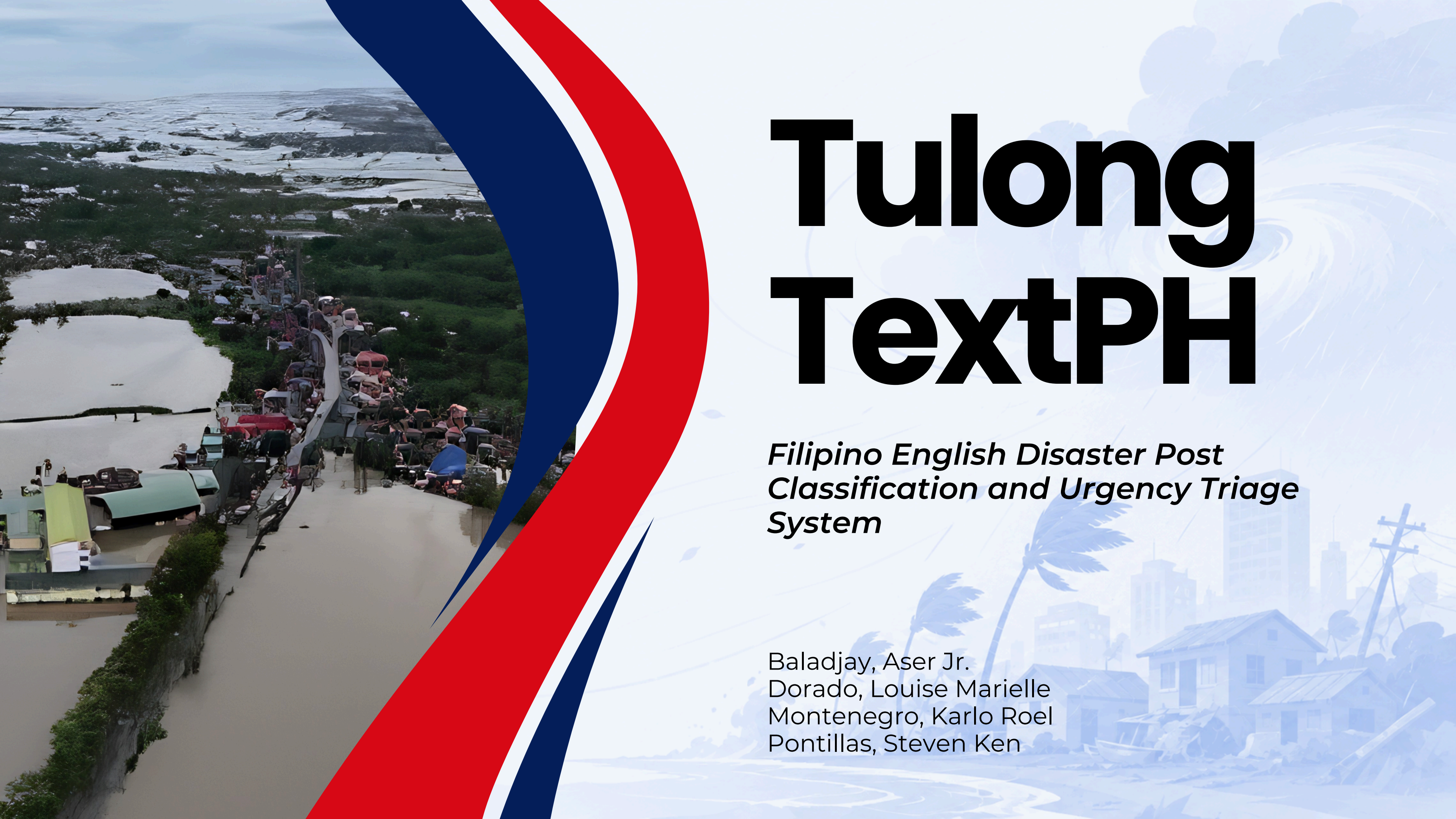




Tulong TextPH

*Filipino English Disaster Post
Classification and Urgency Triage
System*

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Problem Statement

Problem

Disaster-related social media posts are:

- Noisy and informal
- Written in English, Filipino, or Taglish
- Time-sensitive during typhoons, floods, and earthquakes
- Mixed with irrelevant or promotional content
- Hard to manually sort at scale

Project Purpose

TulongText PH automatically classifies disaster-related posts into humanitarian categories and urgency levels to support faster information triage.



LGU



Response
Teams



Volunteers



Monitoring
Teams

Example

Sample Post (Input)

Need rescue sa Brgy. San Pablo, baha na hanggang tuhod!



Predicted Output

📌 **Category:** Rescue or Urgent Needs

🔔 **Urgency:** Critical

Web App Implementation

 **TulongText PH** Home



Filipino-English Disaster Relief Post Classification for Humanitarian Triage

AI-powered classification of social media and disaster posts to help responders prioritize urgent needs and deliver aid faster.

1 Enter Disaster or Social Media Post

Type or paste a disaster or social media post here...

Example: Need rescue sa Brgy. San Isidro, baha na hanggang bubong. May bata at senior na stranded.

0/5000

Model

tfidf_word_bigram_calibrated

Analyze

Clear

2 Analysis Results

Prediction appears here

Run an analysis to view the category, urgency, confidence, cleaned text, and top predictions.

Demo Examples

 Need rescue sa Brgy. San Isidro, baha na hanggang...

 Two injured residents near the collapsed bridge...

 Evacuation center at City High School is open. Bring water,...

 Power lines are down along Mabini Street after the...

 Selling raincoats and flashlights at discounted prices...

Prediction Categories

 Critical	 Critical	 High	 High	 Moderate	 Moderate	 Low	 Low
Rescue or Urgent Needs	Medical or Casualties	Evacuation or Displacement	Infrastructure Damage	Warnings or Advice	Donation or Volunteering	General Update	Not Humanitarian
Posts asking for rescue, urgent help, or stranded-person response.	Injuries, deaths, missing people, medical support, or health concerns.	Evacuation centers, displaced residents, shelters, and relocation needs.	Damaged roads, bridges, power lines, buildings, landslides, or utilities.	Alerts, advisories, caution messages, and safety instructions.	Relief goods, donations, volunteer calls, supplies, food, or water offers.	Relevant disaster updates, reactions, support, or non-urgent information.	Posts that are unrelated, promotional, commercial, or not useful for triage.



Frontend

HTML, CSS, JavaScript



Backend

Python, Flask, Flask-CORS



API Endpoints

GET /api/health

GET /api/models

POST /api/predict

POST /api/batch_predict



The web app makes the NLP model accessible through a simple browser-based interface.

Conclusion, Limitations & Q&A



Key Takeaway

RescueText PH helps organize noisy disaster-related posts into actionable humanitarian categories and urgency levels.



Current Limitations

- Urgency is currently rule-based
- Location extraction is not yet implemented
- Taglish and Filipino labeled data is still limited



Future Improvements

- Add location extraction
- Create a batch triage dashboard
- Add CSV export and explainability
- Train the transformer model further